

Institute of Business Administration – Mid-Term Exam

PART B: Subjective – 90 minutes / 35 marks

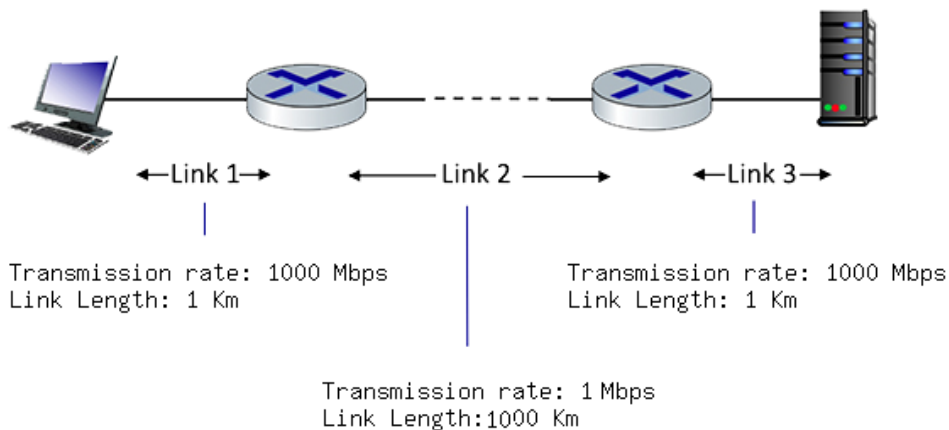
For official use only:

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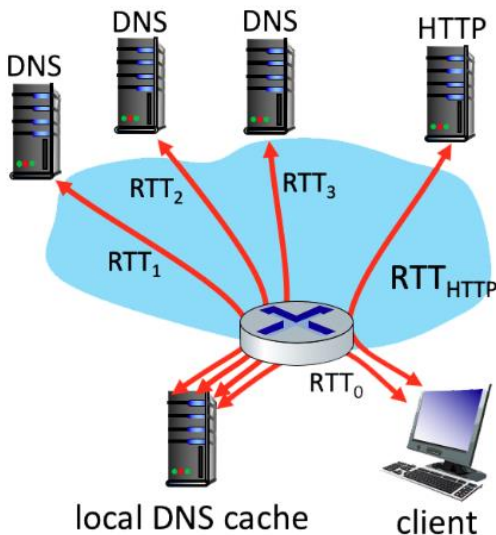
Course: CSE248 - Computer Communications & Networking	Exam: Mid Term
Instructors: Asma Touqir / Tasbiha Fatima / Waseem Arain	Total Time: 90 Minutes
Class: BSCS	Date: 9 th March, 2024
Instructions	
- Attempt all questions - Use only the space provided here to answer, so write carefully and legibly. - Understanding the question paper is part of the exam, so no explanations will be offered on questions.	

1. Consider the figure below with three links, each with the specified transmission rate and link length.



Assume the length of a packet is 16000 bits. The speed of light propagation delay on each link is 3×10^8 m/sec. Calculate the total delay? (4)

2. Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Examine the topology and suppose that four DNS servers, as shown below, are visited before your host receives the IP address from DNS. The first DNS server visited is the local DNS cache, with an RTT delay of $RTT_0 = 5$ msec. The second, third and fourth DNS servers contacted have RTTs of 43, 24, and 31 msec, respectively. Initially, let's suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Suppose the RTT between the local host and the Web server containing the object is $RTT_{HTTP} = 22$ msec. (5)



- a) Assuming zero transmission time for the HTML object, how much time (in msec) elapses from when the client clicks on the link until the client receives the object?
-
- b) Now suppose the HTML object references 4 very small objects on the same server. Neglecting transmission times, how much time (in msec) elapses from when the client clicks on the link until the base object and all 4 additional objects are received from web server at the client, assuming non-persistent HTTP and no parallel TCP connections?
-
- c) Suppose the HTML object references 4 very small objects on the same server, but assume that the client is configured to support a maximum of 5 parallel TCP connections, with non-persistent HTTP.
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- d) Suppose the HTML object references 4 very small objects on the same server, but assume that the client is configured to support a maximum of 5 parallel TCP connections, with persistent HTTP.
-
- e) What's the fastest method we've explored: Non-persistent-serial, Non-persistent-parallel, or Persistent-parallel?
-

3. Is it possible for an organization's Web server and mail server to have exactly the same alias for a hostname (for example, Google.com)? What would be the type for the RR that contains the hostname of the mail server? (3)

4. Consider a short, 10-meter link, over which a sender can transmit at a rate of 150 bits/sec in both directions. Suppose that packets containing data are 100,000 bits long, and packets containing only control (e.g., ACK or handshaking) are 200 bits long. Assume that N parallel connections each get $1/N$ of the link bandwidth. Now consider the HTTP protocol and suppose that each downloaded object is 100 Kbits long, and that the initial downloaded object contains 10 referenced objects from the same sender. (4)

a) Would parallel downloads via parallel instances of non-persistent HTTP make sense in this case?

b) Now consider persistent HTTP. Do you expect significant gains over the non-persistent case? Justify and explain your answer.

5. Suppose you would like to urgently deliver 50 terabytes data from Boston to Los Angeles. You have available a 200 Mbps dedicated link for data transfer. Would you prefer to transmit the data via this link or instead use FedEx overnight delivery? Explain. (4)

-
-
-
6. The text below shows the reply sent from the server in response to the HTTP GET message in the question above. Answer the following questions, indicating where in the message below you find the answer. (4)

```
HTTP/1.1 200 OK<cr><lf>Date: Tue, 07 Mar 2008
12:39:45GMT<cr><lf>Server: Apache/2.0.52 (Fedora)
<cr><lf>Last-Modified: Sat, 10 Dec2005 18:27:46
GMT<cr><lf>ETag: "526c3-f22-a88a4c80"<cr><lf>Accept-
Ranges: bytes<cr><lf>Content-Length: 3874<cr><lf>
Keep-Alive: timeout=max=100<cr><lf>Connection:
Keep-Alive<cr><lf>Content-Type: text/html; charset=
ISO-8859-1<cr><lf><cr><lf><!doctype html public "-
//w3c//dtd html 4.0transitional//en"><lf><html><lf>
<head><lf> <meta http-equiv="Content-Type"
content="text/html; charset=iso-8859-1"><lf> <meta
name="GENERATOR" content="Mozilla/4.79 [en] (Windows NT
5.0; U) Netscape]"><lf> <title>CMPSCI 453 / 591 /
NTU-ST550ASpring 2005 homepage</title><lf></head><lf>
<much more document text following here (not shown)>
```

- a. Was the server able to successfully find the document or not? What time was the document reply provided?

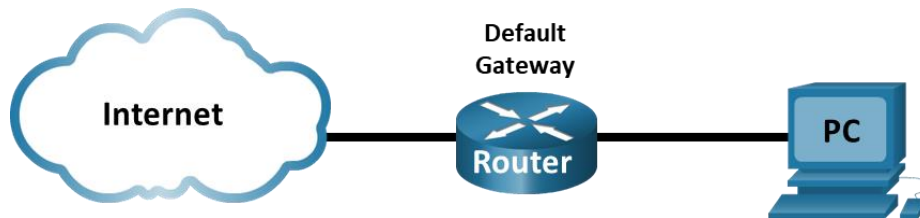
-
- b. When was the document last modified?

-
- c. How many bytes are there in the document being returned?

-
- d. What are the first 5 bytes of the document being returned? Did the server agree to a persistent connection?
-

7. Consider the following topology:

(7)



PC's cmd prompt yields the following output on execution of ipconfig command

```

C:\> ipconfig /all

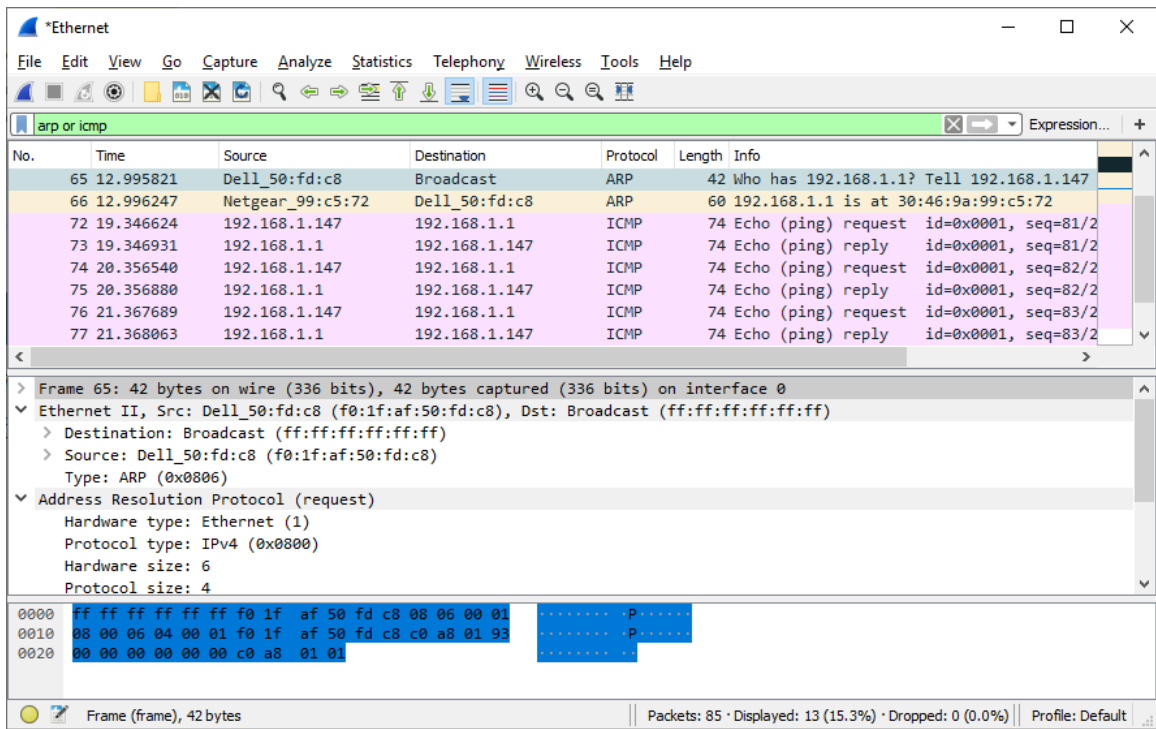
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) 82579LM Gigabit Network Connection
    Physical Address. . . . . : F0-1F-AF-50-FD-C8
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::58c5:45f2:7e5e:29c2%11(Preferred)
    IPv4 Address. . . . . : 192.168.1.147(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Friday, September 6, 2019 11:08:36 AM
    Lease Expires . . . . . : Saturday, September 7, 2019 11:08:36 AM
    Default Gateway . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    <output omitted>

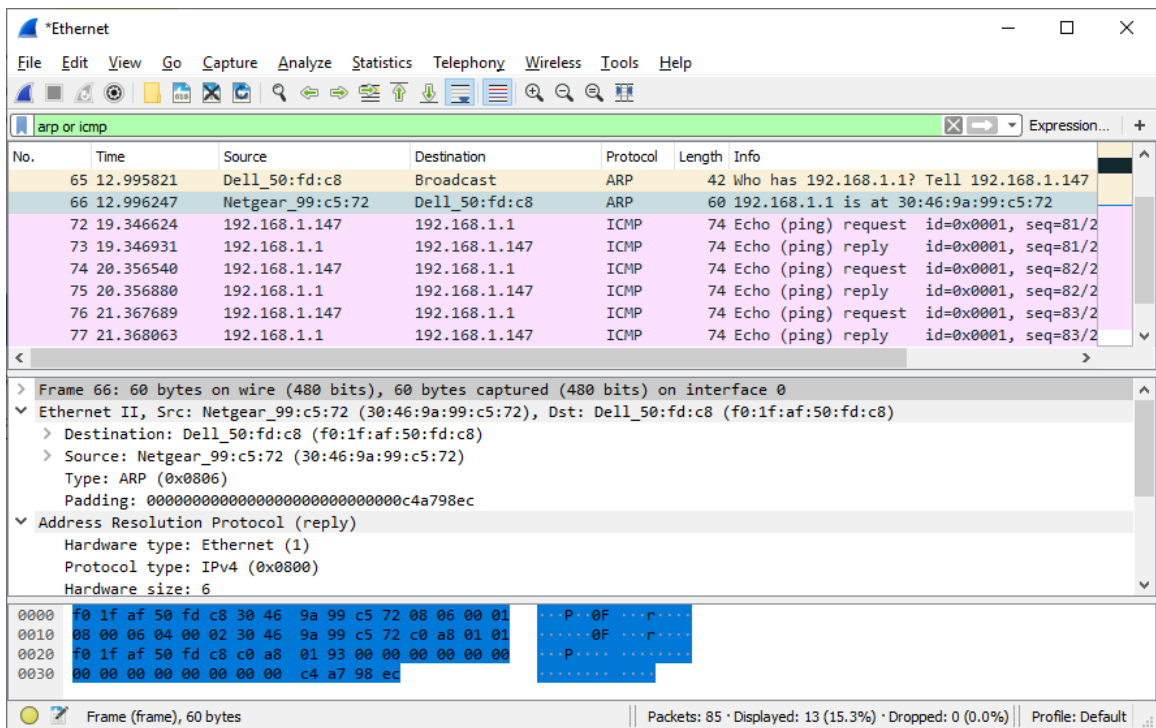
```

The screenshots of the Wireshark capture below show the packets generated by a ping being issued from a PC host to its default gateway. A filter has been applied to Wireshark to view the ARP and ICMP protocols only. ARP stands for address resolution protocol. ARP is a communication protocol that is used for determining the MAC address that is associated with the IP address. The session begins with an ARP query and reply for the MAC address of the gateway router, followed by four ping requests and replies.

This screenshot highlights the frame details for an ARP request.



This screenshot highlights the frame details for an ARP reply.



Now examine the ethernet II header contents of the ARP request/reply and answer the following.

a. What is significant about the contents of the destination address field?

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b. Why does the PC send out a broadcast ARP prior to sending the first ping request?

c. What is the MAC address of the source in the first frame?

d. What is the Vendor ID (OUI) of the Source NIC in the ARP reply?

e. What portion of the MAC address is the OUI?

f. What is the NIC serial number of the source?

g. Wireshark does not display the preamble field of a frame header. What does the preamble contain?

8. Following is the output of *show mac address-table* command on a switch

(4)

```
S2# show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
All     0100.0ccc.cccc   STATIC    CPU
All     0100.0ccc.cccd   STATIC    CPU
...
All     0180.c200.000c   STATIC    CPU
All     0180.c200.000d   STATIC    CPU
All     0180.c200.000e   STATIC    CPU
All     0180.c200.000f   STATIC    CPU
All     0180.c200.0010   STATIC    CPU
All     ffff.ffff.ffff   STATIC    CPU
1       001a.e3cf.b883   DYNAMIC   Fa0/1
1       001a.e3cf.b8c0   DYNAMIC   Fa0/1
1       0050.56b3.27d6   DYNAMIC   Fa0/1
1       0050.56b3.ff54   DYNAMIC   Fa0/18
Total Mac Addresses for this criterion: 24
```

- a. What MAC addresses are recorded in the table? To which switch ports are they mapped and to which devices do they belong? Ignore MAC addresses that are mapped to the CPU.

- b. Why do you see more than one MAC address mapped onto a single port above?
